First- and Second-Order Coherence Through Spacetime Foam

Author One
Department of Physics, University X, City, Country
(Dated: April 22, 2026)

We study first- and second-order coherence of light from point and extended sources propagating through spacetime foam characterized by the Ng–Perlman exponent α. In the narrow-band regime, arrival-time jitter suppresses temporal first-order coherence and produces a Siegert violation whose foam dependence encodes α and the line-of-sight path-length scale $\sigma_\ell(D)$. For extended sources, transverse foam coherence introduces an additional suppression of the zero-lag Hanbury Brown–Twiss bunching peak that directly probes the transverse coherence length $r_\perp$. Joint measurements of the Siegert discrepancy and zero-lag bunching suppression across sources of different angular size and redshift can constrain $\{\alpha, \sigma_\ell(D), r_\perp\}$ simultaneously.

I. MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{1}$$

where $E_{\rm src}(t)$ is a stationary zero-mean complex Gaussian process (a single chaotic/thermal mode) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of $E_{\rm src}$. The jitter arises from cumulative path-length fluctuations along the line of sight of length $\ell^{(D)}$:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell^{(D)}_{\rm tot}(t). \tag{2}$$

B. Ng–Perlman Variance and Temporal Covariance

Following Ng and collaborators [1–3], the single-ray path-length variance obeys the power-law scaling

$$\sigma^2_\ell(D) = A_\alpha\,[\ell^{(D)}]^{2(1-\alpha)}\,\ell_{\rm P}^{2\alpha}, \tag{3}$$

where $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616\times10^{-35}$ m is the Planck length, α parametrizes the foam model ($\alpha = 1$: conservative; $\alpha = 2/3$: holographic; $\alpha = 1/2$: random walk), and A_α is a dimensionless coefficient of order unity. The temporal covariance of the aggregate path-length fluctuation along a single ray takes the factorized form

$$\left\langle \Delta\ell^{(D)}_{\rm tot}(t)\,\Delta\ell^{(D)}_{\rm tot}(t+\tau)\right\rangle = \sigma^2_\ell(D)\,C_\alpha(\tau; D), \tag{4}$$

where the temporal decorrelation function $C_\alpha(\tau; D)$ satisfies $C_\alpha(0; D) = 1$ and decays to zero as $\tau \to \infty$.

C. Differential Jitter and Its Distribution

Define the differential jitter between two detection times separated by lag τ:

$$\Delta(\tau) \equiv \delta t(t+\tau) - \delta t(t). \tag{5}$$

For a fixed propagation distance D, stationarity implies that the statistics of $\Delta(\tau)$ depend only on τ. Its variance follows directly from Eq. (4):

$$\begin{aligned}\sigma^2_\Delta(\tau; D) &\equiv \langle \Delta(\tau)^2\rangle \\ &= \frac{1}{c^2}\left\langle [\Delta\ell_{\rm tot}(t+\tau) - \Delta\ell_{\rm tot}(t)]^2\right\rangle \\ &= \frac{2\sigma^2_\ell(D)}{c^2}\big[1 - C_\alpha(\tau; D)\big].\end{aligned} \tag{6}$$

Because $\Delta(\tau)$ is a sum of contributions from many weakly correlated foam cells, the central limit theorem implies it is Gaussian with zero mean. We therefore write its conditional probability density (at fixed D) as

$$p(\Delta;\tau,D) = \frac{1}{\sqrt{2\pi}\,\sigma_\Delta(\tau;D)} \exp\left(-\frac{\Delta^2}{2\sigma_\Delta^2(\tau;D)}\right). \tag{7}$$

Note the two limits: $\sigma_\Delta^2(0;D) = 0$ (no jitter between simultaneous arrivals) and $\sigma_\Delta^2(\infty;D) = 2\sigma_\ell^2(D)/c^2$ (maximal decorrelation at large lag).

For notational compactness in subsequent sections, we suppress the explicit D-dependence and write $C_\alpha(\tau) \equiv C_\alpha(\tau;D)$ and $\sigma_\Delta(\tau) \equiv \sigma_\Delta(\tau;D)$ (similarly for any other D-dependent quantity).

II. FIRST-ORDER COHERENCE

A. General Result

The observed first-order coherence is

$$g^{(1)}_{\rm obs}(\tau) = \frac{\langle E^*_{\rm obs}(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle |E_{\rm obs}(t)|^2\rangle}. \tag{8}$$

Substituting Eq. (1) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E^*_{\rm src}(u)\, E_{\rm src}(v)\rangle = \langle |E_{\rm src}|^2\rangle\, g^{(1)}_{\rm src}(\tau - \Delta(\tau)). \tag{9}$$

Since δt is independent of $E_{\rm src}$, averaging over the jitter yields

$$\boxed{g^{(1)}_{\rm obs}(\tau) = \left\langle g^{(1)}_{\rm src}(\tau - \Delta(\tau))\right\rangle_{\delta t} = \int_{-\infty}^{\infty} p(\Delta;\tau)\, g^{(1)}_{\rm src}(\tau - \Delta)\, d\Delta.} \tag{10}$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

B. Closed Form for a Lorentzian Source

A thermal or AGN source typically has a Lorentzian power spectrum, so

$$g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_c}, \tag{11}$$

where τ_c is the coherence time. Evaluating Eq. (10) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g^{(1)}_{\rm obs}(\tau) = e^{\sigma_\Delta^2/2\tau_c^2} \cdot \frac{1}{2}\left[e^{-\tau/\tau_c}\,{\rm erfc}\left(\frac{\sigma_\Delta/\tau_c - \tau/\sigma_\Delta}{\sqrt{2}}\right) + e^{\tau/\tau_c}\,{\rm erfc}\left(\frac{\sigma_\Delta/\tau_c + \tau/\sigma_\Delta}{\sqrt{2}}\right)\right], \tag{12}$$

where $\sigma_\Delta \equiv \sigma_\Delta(\tau)$. In the weak-jitter limit $\sigma_\Delta \ll \tau_c$, the Gaussian average of $e^{-|\tau-\Delta|/\tau_c}$ over $\Delta \sim \mathcal{N}(0,\sigma_\Delta^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g^{(1)}_{\rm obs}(\tau) \approx e^{-|\tau|/\tau_c}\left[1 + \frac{\sigma_\Delta^2(\tau)}{2\tau_c^2} + \mathcal{O}(\sigma_\Delta^4)\right]. \tag{13}$$

The correction is positive because $e^{-|\tau|/\tau_c}$ is convex and Jensen's inequality applies. The normalization $g^{(1)}_{\rm obs}(0) = 1$ is preserved exactly, since $\sigma_\Delta(0) = 0$.

C. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, write $E_{\rm src}(t) = A(t)\, e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c\,\Delta\lambda)$. For jitter $|\delta t| \ll \tau_c$, the envelope is unaffected, $A(t-\delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation:

$$E_{\rm obs}(t) \approx A(t)\, e^{-i\omega_0 t}\, e^{i\omega_0 \delta t(t)}. \tag{14}$$

Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of the Gaussian $\Delta(\tau)$ evaluated at ω_0:

$$\left\langle e^{i\omega_0 \Delta(\tau)} \right\rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \tag{15}$$

Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (6):

$$\boxed{g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \exp\!\left(-\sigma_\phi^2 \left[1 - C_\alpha(\tau; D)\right]\right),} \tag{16}$$

where we define the rms accumulated phase

$$\sigma_\phi \equiv \frac{2\pi\, \sigma_\ell(D)}{\lambda} = \frac{2\pi\, \sqrt{A_\alpha}\, [\ell^{(D)}]^{1-\alpha}\, \ell_{\rm P}^{\alpha}}{\lambda}. \tag{17}$$

We write the foam suppression factor as

$$\mathcal{F}_{\rm foam}(\tau) \equiv e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \tag{18}$$

so $g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\rm foam}(0) = 1$. At $\tau \to \infty$: $\mathcal{F}_{\rm foam}(\infty) = e^{-\sigma_\phi^2}$.

D. Connection to Fringe Visibility and Strehl Ratio

1. *Fringe Visibility and the Lieu–Hillman Condition*

The fringe visibility measured in a stellar interferometer at time delay τ is $V(\tau) = |g^{(1)}_{\rm obs}(\tau)|$. From Eq. (16):

$$V(\tau) = V_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau), \tag{19}$$

where $V_{\rm src}(\tau) = |g^{(1)}_{\rm src}(\tau)|$ is the intrinsic source visibility. Lieu & Hillman [4] argue that fringes are erased when the accumulated phase randomizes over $[0, 2\pi)$, i.e. $\sigma_\phi \gtrsim 1$. In our language this is the limit $\mathcal{F}_{\rm foam} \to 0$: the foam suppression factor collapses and $V \to 0$. The Lieu–Hillman condition $\Delta\phi \geq 2\pi$, where $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$ [1], is precisely $\sigma_\phi \geq 1$. Our result extends this binary criterion to a continuous suppression curve $V(\tau)$ whose shape encodes the temporal decorrelation $C_\alpha(\tau; D)$.

2. *Strehl Ratio*

For spatially uncorrelated phase noise of rms σ_ϕ across the telescope aperture, the Strehl ratio (ratio of the observed PSF peak to the diffraction-limited peak) is [1]

$$S = e^{-\sigma_\phi^2}. \tag{20}$$

Comparing with Eq. (18), the Strehl ratio equals the long-lag limit of the foam suppression factor:

$$S = \mathcal{F}_{\rm foam}(\tau \to \infty). \tag{21}$$

Both the Strehl and the Lieu–Hillman fringe visibility measure the same underlying quantity $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ — the total accumulated phase variance. The difference is observational channel: Strehl and fringes probe the *spatial* phase variation across the aperture, while the temporal $g^{(1)}$ probes the *temporal* phase variation along a single ray as a function of lag.

III. SECOND-ORDER COHERENCE

A. Conditional Siegert Relation

Fix a realization of the jitter process $\delta t(\cdot)$. Conditioned on it, $E_{\rm obs}(t)$ is a deterministic time-warp of a complex Gaussian field, hence is itself complex Gaussian. For any complex Gaussian field the Siegert relation holds exactly, so

$$g^{(2)}_{\rm obs}(\tau \mid \delta t) - 1 = \left| g^{(1)}_{\rm obs}(\tau \mid \delta t) \right|^2 . \tag{22}$$

Given a fixed δt, the differential jitter $\Delta(\tau)$ is a deterministic number, so $g^{(1)}_{\rm obs}(\tau \mid \delta t) = g^{(1)}_{\rm src}(\tau - \Delta(\tau))$. Using the intrinsic Siegert relation $g^{(2)}_{\rm src} - 1 = |g^{(1)}_{\rm src}|^2$:

$$g^{(2)}_{\rm obs}(\tau \mid \delta t) - 1 = \left| g^{(1)}_{\rm src}(\tau - \Delta(\tau)) \right|^2 = g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1. \tag{23}$$

B. Unconditional Second-Order Coherence

Averaging over the jitter distribution:

$$\boxed{g^{(2)}_{\rm obs}(\tau) - 1 = \left\langle g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1 \right\rangle_{\delta t} = \int_{-\infty}^{\infty} p(\Delta;\tau) \left[g^{(2)}_{\rm src}(\tau - \Delta) - 1 \right] d\Delta.} \tag{24}$$

The structure is identical to Eq. (10): both $g^{(1)}_{\rm obs}$ and $g^{(2)}_{\rm obs}$ are convolutions of their intrinsic counterparts with the *same* Gaussian jitter kernel.

C. Narrow-Band Limit: $g^{(2)}$ is Unchanged to Leading Order

In the narrow-band limit, the jitter $\delta t \ll \tau_c$ does not affect the amplitude envelope $A(t)$ appreciably. Since the field is $E_{\rm obs}(t) = A(t)\, e^{-i\omega_0 t}\, e^{i\omega_0 \delta t(t)}$ (with $A(t - \delta t) \approx A(t)$ for slow envelope variations), the intensity is

$$I_{\rm obs}(t) = |E_{\rm obs}(t)|^2 = |A(t - \delta t(t))|^2. \tag{25}$$

In the limit $\delta t/\tau_c \to 0$ we may set $A(t - \delta t) \approx A(t)$, so that to leading order

$$I_{\rm obs}(t) \approx |A(t)|^2 = I_{\rm src}(t). \tag{26}$$

Consequently, the intensity statistics agree with the intrinsic source to leading order in the narrow-band expansion:

$$g^{(2)}_{\rm obs}(\tau) = g^{(2)}_{\rm src}(\tau) \qquad \text{(narrow-band limit, to leading order)}, \tag{27}$$

with corrections controlled by the small envelope-time shifts (equivalently by $\sigma_\Delta(\tau)/\tau_c$).

The physical reason is that intensity is blind to the carrier phase $e^{i\omega_0 \delta t(t)}$: carrier-phase modulation does not affect $|E_{\rm obs}(t)|^2$. By contrast, $g^{(1)}$ *is* sensitive to this phase because $\omega_0 \delta t \sim \omega_0 \sigma_\ell / c = \sigma_\phi$ can be large even when $\delta t \ll \tau_c$: for $\alpha = 1/2$ at $D = 1$ Gpc and $\lambda = 500$ nm, $\sigma_\phi \approx 280 \gg 1$.

D. Why Temporal Second-Order Correlations are Less Sensitive to First-Order Effects

Although the narrow-band jitter leaves $g^{(2)}$ for a point source unchanged at leading order, any observable foam signature in a temporal second-order measurement is controlled by the residual differential-jitter scale $\sigma_\Delta(\tau)$ relative to the source coherence time τ_c. In the narrow-band limit the relevant sensitivity parameter for the temporal measurement is $\sigma_\Delta(\tau)/\tau_c$, not σ_ϕ directly. Using $c\tau_c = \lambda^2/(2\pi\Delta\lambda)$ (physics convention) and $\omega_0 = 2\pi c/\lambda$, so $\omega_0 \tau_c = \lambda/\Delta\lambda$:

$$\frac{\sigma_\Delta(\tau)}{\tau_c} = \frac{\sqrt{2[1 - C_\alpha(\tau)]}\, \sigma_\ell(D)}{c\tau_c} = \frac{\sigma_\phi}{\omega_0 \tau_c} \cdot \sqrt{2[1 - C_\alpha(\tau)]} = \sigma_\phi \cdot \frac{\Delta\lambda}{\lambda} \cdot \sqrt{2[1 - C_\alpha(\tau)]}. \tag{28}$$

TABLE I. Comparison of foam observables at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$.

Method	Probe	Sensitivity
Strehl / Lieu–Hillman	$\sigma_\phi = 2\pi\sigma_\ell/\lambda$	Direct
Temporal $g^{(1)}$ at lag τ	$\sigma_\phi\sqrt{2[1 - C_\alpha(\tau)]} \cdot \Delta\lambda/\lambda$	Suppressed by $\Delta\lambda/\lambda$

The suppression factor $\Delta\lambda/\lambda \ll 1$ means that even when $\sigma_\phi \gg 1$ (fringes erased), the temporal HBT ratio can be of order unity or smaller.

For $\alpha = 1/2$ at $D = 1$ Gpc with $A_\alpha = 1$: $\sigma_\ell = \sqrt{\ell_{\rm P}\,\ell^{(D)}} \approx 2.2 \times 10^{-5}$ m, giving $\sigma_\phi \approx 280$ (fringes completely erased at $\sigma_\phi \gg 1$), while $c\tau_c = \lambda^2/(2\pi\Delta\lambda) \approx 80\,\mu$m and $\sigma_\Delta/\tau_c \approx \sqrt{2}\,\sigma_\ell/(c\tau_c) \approx 0.40$ (marginally accessible to HBT). This reflects the fact that phase-coherence constraints already rule out $\alpha = 1/2$ at optical wavelengths [5], whereas the temporal HBT signature for the same model is only at the edge of detectability.

IV. SIEGERT RELATION VIOLATION

A. General Bound

A foundational result for thermal/chaotic light is the Siegert relation, which connects the second-order (intensity–intensity) coherence to the first-order (field–field) coherence: for stationary complex Gaussian fields one has $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$. In practice, this means that once $g^{(1)}$ is known, the expected $g^{(2)}$ is fixed, providing a powerful internal consistency check on the underlying statistical assumptions. Systems that do not follow the Siegert relation are therefore of special interest: they signal the presence of nontrivial propagation or measurement physics—e.g., non-Gaussianity, mode mixing, or additional stochastic phase/amplitude processes—that can decouple intensity correlations from field coherence. Such Siegert violations offer a route to probe otherwise hidden dynamics with a self-referential observable, making them particularly valuable in contexts where conventional first-order coherence tests are insensitive or ambiguous.

Define the Siegert discrepancy:

$$\mathcal{S}(\tau) \equiv \left[g^{(2)}_{\rm obs}(\tau) - 1\right] - \left|g^{(1)}_{\rm obs}(\tau)\right|^2. \tag{29}$$

Let $X \equiv g^{(1)}_{\rm src}(\tau - \Delta(\tau))$, a complex random variable. From Eqs. (10) and (24):

$$g^{(1)}_{\rm obs}(\tau) = \langle X\rangle_{\delta t}, \qquad g^{(2)}_{\rm obs}(\tau) - 1 = \langle |X|^2\rangle_{\delta t}. \tag{30}$$

Therefore

$$\mathcal{S}(\tau) = \langle |X|^2\rangle - |\langle X\rangle|^2 \geq 0, \tag{31}$$

by the Cauchy–Schwarz inequality, with equality only when $\Delta(\tau)$ has zero variance (i.e., at $\tau = 0$ or in the absence of foam). The Siegert relation is violated for all $\tau \neq 0$ whenever foam is present. This non-negativity holds for a point source; for extended sources the cross-ray differential jitter can drive $\mathcal{S}(0) < 0$ (see Section V).

B. Explicit Form in the Narrow-Band Limit

In the narrow-band limit, $g^{(2)}_{\rm obs} \approx g^{(2)}_{\rm src}$ to leading order (from Eq. (27)), while $g^{(1)}_{\rm obs}$ is suppressed by $F_{\rm foam}(\tau)$ (Eq. (**??**)). Using the intrinsic Siegert relation $g^{(2)}_{\rm src}(\tau) - 1 = |g^{(1)}_{\rm src}(\tau)|^2$, to leading order

$$S(\tau) \approx |g^{(1)}_{\rm src}(\tau)|^2\left(1 - e^{-2\sigma_\phi^2[1 - C_\alpha(\tau;D)]}\right).$$

C. Properties

The key properties of $\mathcal{S}(\tau)$ are summarized in Table II.

TABLE II. Properties of the Siegert discrepancy $\mathcal{S}(\tau)$ in the narrow-band limit.

Limit	$\mathcal{S}(\tau)$	Reason
$\tau = 0$	0	$\Delta(0) = 0$; both coherences agree
$\sigma_\phi \ll 1$ (weak jitter)	$\approx 2\sigma_\phi^2[1 - C_\alpha]\,\lvert g^{(1)}_{\rm src}\rvert^2$	Linear in phase variance
$\sigma_\phi \gg 1$ (strong jitter)	$\approx \lvert g^{(1)}_{\rm src}(\tau)\rvert^2$	$g^{(1)}_{\rm obs} \to 0$; full violation
$\tau \to \infty$	$\to 0$	$\lvert g^{(1)}_{\rm src}\rvert \to 0$ faster

D. Physical Interpretation

In the narrow-band limit foam acts as a *phase screen*: it modulates the carrier phase $\omega_0 \delta t(t)$ randomly without touching the intensity envelope. This asymmetry is the root cause of the Siegert violation. The discrepancy $\mathcal{S}(\tau)$ is a direct observable — both $g^{(1)}_{\rm obs}$ and $g^{(2)}_{\rm obs}$ are measurable from the same dataset — and carries a specific distance dependence $\sigma_\phi^2 \propto [\ell^{(D)}]^{2(1-\alpha)}$ that distinguishes foam from instrumental effects, which would not scale with source distance. The modulation of $\mathcal{S}(\tau)$ beyond the prefactor $\lvert g^{(1)}_{\rm src}(\tau)\rvert^2$ is controlled solely by α, D, and $C_\alpha(\tau; D)$, independently of the unknown source spectrum.

V. EXTENDED SOURCES: TRANSVERSE FOAM COHERENCE

A. Field Model and Spatiotemporal Covariance

We now generalize to an extended incoherent source whose angular brightness distribution is $W(\boldsymbol{\theta})$ (normalized to unity). The detected field is a weighted superposition over rays labelled by direction offset $\boldsymbol{\theta}$:

$$E_{\rm obs}(t) = \int d\boldsymbol{\theta}\, W^{1/2}(\boldsymbol{\theta})\, E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta}))\,, \tag{32}$$

where the $E_{\boldsymbol{\theta}}$ are independent, identically distributed complex Gaussian processes (one per sky direction), and each ray carries its own foam jitter $\delta t(t; \boldsymbol{\theta})$. The spatiotemporal covariance of the path-length aggregate is

$$\big\langle \Delta\ell_{\rm tot}(t; \boldsymbol{\theta})\, \Delta\ell_{\rm tot}(t + \tau; \boldsymbol{\theta}')\big\rangle = \sigma_\ell^2(D)\, C_\alpha(\tau; D)\, \Phi\!\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \tag{33}$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq \ell^{(D)}|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ is the transverse separation between rays at distance $\ell^{(D)}$, $r_\perp$ is the foam transverse coherence length, and Φ is the transverse coherence function satisfying $\Phi(0) = 1$ and $\Phi(b/r_\perp) \to 0$ for $b \gg r_\perp$.

B. First-Order Coherence is Unchanged

The numerator of $g^{(1)}_{\rm obs}(\tau)$ involves cross terms $\langle E^*_{\boldsymbol{\theta}}(\ldots)\, E_{\boldsymbol{\theta}'}(\ldots)\rangle$. Because different source directions are spatially incoherent,

$$\langle E^*_{\boldsymbol{\theta}}(s)\, E_{\boldsymbol{\theta}'}(s')\rangle \propto \delta(\boldsymbol{\theta} - \boldsymbol{\theta}'), \tag{34}$$

so only the diagonal $\boldsymbol{\theta} = \boldsymbol{\theta}'$ contributes. Each diagonal term reproduces the single-ray result of Section II C, and the angular integral over W gives unity. Therefore:

$$g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot e^{-\sigma_\phi^2[1 - C_\alpha(\tau; D)]}, \tag{35}$$

identical to the point-source result Eq. (16). The first-order coherence probes only single-ray statistics; the transverse foam structure leaves no imprint on $g^{(1)}$.

C. Cross-Ray Differential Jitter

For $g^{(2)}$, the Wick contraction in the Gaussian moment theorem enforces pairings between rays $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ that need not be equal. The relevant quantity is the *cross-ray differential jitter*:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \equiv \delta t(t+\tau;\boldsymbol{\theta}) - \delta t(t;\boldsymbol{\theta}'), \tag{36}$$

the time offset between the "second photon" arriving along $\boldsymbol{\theta}$ at $t+\tau$ and the "first photon" along $\boldsymbol{\theta}'$ at t. Computing its variance from Eq. (33):

$$\begin{aligned}\sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}') &= \frac{1}{c^2}\big\langle[\Delta\ell_{\rm tot}(t+\tau;\boldsymbol{\theta}) - \Delta\ell_{\rm tot}(t;\boldsymbol{\theta}')]^2\big\rangle \\ &= \frac{1}{c^2}\left[\sigma^2_\ell + \sigma^2_\ell - 2\sigma^2_\ell\, C_\alpha(\tau)\,\Phi\!\left(\frac{b}{r_\perp}\right)\right] \\ &= \frac{2\sigma^2_\ell(D)}{c^2}\left[1 - C_\alpha(\tau;D)\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right)\right].\end{aligned} \tag{37}$$

Setting $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (and hence $\Phi(0) = 1$) recovers the single-ray result Eq. (6) exactly. For $b \gg r_\perp$, $\Phi \to 0$ and the variance becomes $2\sigma^2_\ell/c^2$ *at all lags, including* $\tau = 0$ — a new effect absent in the point-source case.

D. General Second-Order Coherence

By the Gaussian moment theorem, the unconditional $g^{(2)}$ for an extended source is an angularly weighted mixture over all ray pairs:

$$\boxed{g^{(2)}_{\rm obs}(\tau) - 1 = \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; W(\boldsymbol{\theta})W(\boldsymbol{\theta}') \int_{-\infty}^{\infty} p_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\Delta;\tau)\left[g^{(2)}_{\rm src}(\tau-\Delta) - 1\right] d\Delta,} \tag{38}$$

where $p_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\Delta;\tau)$ is Gaussian with variance $\sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')$ from Eq. (37). In the limit $\Phi \equiv 1$ (all rays transversely coherent), every kernel reduces to the single-ray $p_{\rm 1D}$ and Eq. (38) reduces to the point-source result Eq. (24).

E. Effective Transverse Coherence Factor

To reduce Eq. (38) to a tractable form, define the effective transverse coherence factor:

$$\Phi_{\rm eff} \equiv \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; W(\boldsymbol{\theta})W(\boldsymbol{\theta}')\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right). \tag{39}$$

This replaces the ray-dependent differential variance with a single effective variance:

$$\sigma^2_{\tau,\rm eff}(\tau) \equiv \frac{2\sigma^2_\ell(D)}{c^2}\Big[1 - C_\alpha(\tau;D)\,\Phi_{\rm eff}\Big], \tag{40}$$

and allows Eq. (38) to be approximated by the simpler 1D convolution of Eq. (24) with σ^2_Δ replaced by $\sigma^2_{\tau,\rm eff}$. The two limits are: $\Phi_{\rm eff} \to 1$ (point-like source or $r_\perp \gg \ell^{(D)}\Delta\theta$) recovers the single-ray result; $\Phi_{\rm eff} \to 0$ (fully incoherent, $r_\perp \ll \ell^{(D)}\Delta\theta$) maximizes the effective variance.

F. Zero-Lag Bunching Peak Suppression

At $\tau = 0$, $C_\alpha(0;D) = 1$, so $\sigma^2_{\tau,\rm eff}(0)$ reduces to

$$\sigma^2_0 \equiv \frac{2\sigma^2_\ell(D)}{c^2}(1 - \Phi_{\rm eff}), \tag{41}$$

which is strictly positive whenever $\Phi_{\rm eff} < 1$. Within the $\Phi_{\rm eff}$ approximation of Section V E, the ray-pair mixture reduces to a single Gaussian kernel of width σ_0. Evaluating the zero-lag convolution using $g^{(2)}_{\rm src}(\tau) - 1 = e^{-2|\tau|/\tau_c}$ (from Siegert applied to $g^{(1)}_{\rm src} = e^{-|\tau|/\tau_c}$): For $\Delta \sim \mathcal{N}(0, \sigma_0^2)$ and $s \equiv \sigma_0/\tau_c$:

$$\left\langle e^{-2|\Delta|/\tau_c} \right\rangle = e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big)\,, \tag{42}$$

so

$$\boxed{g^{(2)}_{\rm obs}(0) = 1 + e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big)\,, \qquad s \equiv \sigma_0/\tau_c,} \tag{43}$$

and the zero-lag bunching peak is suppressed below its ideal value of 2 by

$$\delta g^{(2)} \equiv 2 - g^{(2)}_{\rm obs}(0) = 1 - e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big)\,. \tag{44}$$

In the point-source limit $\Phi_{\rm eff} \to 1$, $\sigma_0 \to 0$ and $\delta g^{(2)} \to 0$ as expected. Since $\sigma_0^2 \propto (1 - \Phi_{\rm eff})$ and $\Phi_{\rm eff}$ decreases as the source angular diameter increases at fixed D and $r_\perp$, more extended sources experience greater suppression of the bunching peak.

The physical origin is distinct from the temporal phase-scrambling effect. Even at $\tau = 0$, photon pairs from *different* sky directions $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ accumulate independent instantaneous jitters $\delta t(t; \boldsymbol{\theta}) \neq \delta t(t; \boldsymbol{\theta}')$ when $b \gg r_\perp$. Their relative offset $\delta t(t; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}')$ blurs the $g^{(2)}(0)$ measurement even though $\tau = 0$. This is a purely *spatial* effect with no analogue in the point-source case.

G. Extended-Source Siegert Discrepancy

For the extended source, $g^{(1)}_{\rm obs}(\tau)$ is unchanged from the point-source result (Eq. 35), while $g^{(2)}_{\rm obs}(\tau)$ is given by the general mixture Eq. (38). Unlike the point-source case, in the narrow-band limit $g^{(2)}_{\rm obs}$ is *not* equal to $g^{(2)}_{\rm src}$: the cross-ray differential jitter modifies the intensity correlation through the envelope timing, even though the carrier phase does not affect individual intensities. The Siegert discrepancy is therefore

$$\mathcal{S}(\tau) = \Big[g^{(2)}_{\rm obs}(\tau) - 1\Big] - \Big|g^{(1)}_{\rm obs}(\tau)\Big|^2\,, \tag{45}$$

where $g^{(2)}_{\rm obs}(\tau) - 1$ is given by Eq. (38) and $|g^{(1)}_{\rm obs}(\tau)|^2 = |g^{(1)}_{\rm src}(\tau)|^2\, e^{-2\sigma_\phi^2[1 - C_\alpha(\tau)]}$. Within the $\Phi_{\rm eff}$ approximation, at $\tau = 0$, using $g^{(1)}_{\rm obs}(0) = 1$ exactly:

$$\mathcal{S}(0) = \Big[g^{(2)}_{\rm obs}(0) - 1\Big] - 1 = e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big) - 1 = -\,\delta g^{(2)} < 0, \tag{46}$$

so within this approximation the Siegert discrepancy at zero lag equals minus the bunching-peak suppression, and is negative for $s > 0$. For a point source ($\Phi_{\rm eff} = 1$, $s = 0$), $\mathcal{S}(0) = 0$, recovering the point-source result.

H. Two Distinct Signatures

The two observable signatures of foam for extended sources are summarized in Table III.

TABLE III. Two foam signatures for an extended source in the narrow-band limit.

Signature	Origin	Non-zero at $\tau = 0$?	Depends on $r_\perp$?
Siegert discrepancy $\mathcal{S}(\tau)$	Temporal + cross-ray jitter	Yes (via $\delta g^{(2)}$; < 0)	Yes — via $\Phi_{\rm eff}$
Bunching suppression $\delta g^{(2)}$	Transverse spatial decorrelation	Yes	Yes — direct probe of $r_\perp$

The two signatures are complementary. For a point-like source, the Siegert discrepancy $\mathcal{S}(\tau)$ is controlled by σ_ϕ^2 and $C_\alpha(\tau; D)$ alone, probing α and D with no sensitivity to $r_\perp$. For an extended source, $\mathcal{S}(\tau)$ acquires additional dependence on $r_\perp$ through the cross-ray differential jitter: at $\tau = 0$ this is captured (within the $\Phi_{\rm eff}$ reduction) by $\mathcal{S}(0) = -\delta g^{(2)}$, which depends on $\Phi_{\rm eff}$ and hence on $r_\perp$. The bunching suppression $\delta g^{(2)}$ is controlled by $\sigma_0^2 \propto \sigma_\ell^2(D)(1 - \Phi_{\rm eff})$, providing a direct handle on $r_\perp$ through the dependence of $\Phi_{\rm eff}$ on source angular size. Joint measurements of point-like and extended sources at the same redshift therefore allow $\{\alpha, \sigma_\ell(D), r_\perp\}$ to be constrained simultaneously and largely independently of the unknown intrinsic source spectrum.

ACKNOWLEDGMENTS

[Acknowledgments to be added.]

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